



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 4

1. Title: Use function types, scope, and closures; apply try-catch; build a palindrome checker.

2. Software/Tools Required: Visual Studio Code (VS Code) Google Chrome/web browser, HTML, JavaScript.

3. Theory:

Function Types: Demonstrates anonymous functions (assigned directly to variables without a name) and arrow functions (concise ES6 => syntax).

Scope: Differentiates between **global variables** (accessible throughout the script) and **local variables** (restricted to the block or function where they are declared).

Closures: Occurs when an inner function retains access to its parent function's scope, preserving state variables (like `count`) even after the outer function finishes executing.

Error Handling: Uses `try...catch` and `throw` statements to validate input types and gracefully handle runtime exceptions without crashing the program.



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Palindrome Algorithm: A string is split into an array of characters, reversed, joined back into a string, and compared against the original input using strict equality (===).

4. Output:

```
1 //Palindrome Checker
2 //Name: Sonal Nakhate PRN: 24070521199
3 //Scope, closure, error handling using try-catch
4
5 //Create anonymous function without a name
6
7 let welcome = function() {
8     console.log("Palindrome Checker Program.");
9 }
10
11 welcome();
12
13 //Global Variable
14 let language = "JavaScript";
15
16 function ScopeDemo() {
17     //Local Variable
18     let topic = "Functions";
19     console.log("Language:", language); //Display global var
```

Topic: Functions
Counter: 1
Counter: 2
madam is a Palindrome.
PS C:\Users\Sonal\Desktop\Documents\SEM V\JS_24070521199> node Exp4/Palindrome.js
Palindrome Checker Program.
Language: JavaScript
Topic: Functions
Counter: 1
Counter: 2
madam is a Palindrome.
PS C:\Users\Sonal\Desktop\Documents\SEM V\JS_24070521199>

5. Result/Conclusion: In this experiment, fundamental JavaScript concepts—including anonymous/arrow functions, scope rules, closures, and `try...catch` error handling—were successfully implemented. Using these techniques alongside array manipulation methods (`split`, `reverse`, `join`), a functional and error-resistant palindrome checker was constructed.